bl-sdk / unrealsdk

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On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.

New issue

Unhandled types in BL4 #74

Open

slippycheeze opened on Mar 2 · edited by slippycheeze

Edits

G'day. There are a few unhandled types in BL4 that frequently show up, especially in structures, which make it more difficult to work with the game than ideal. The ones I can currently identify are:

`struct FGbxDef`, AKA `/Script/GbxNexus.GbxDef`, which isn't so much unhandled in the sense the others are, but rather, is an opaque blob of memory that all the Nexus stuff is expressed in, and which at the least probably needs to be copyable to be able to work with a bunch of other structs. Find them everywhere, except for a query interface exposed via UHT. :)

`struct FGbxInlineStruct`, which is used all over the place, and can't be accessed or (importantly) copied, which makes working with ... I want to say "most" structures difficult. Weirdly, all the places it is used strongly suggest that it stays 0x18 bytes in size all the time, so I don't know — I'd guess it was some variable size thing, but that doesn't make any sense if it has other objects immediately after.

The `TestInlineStructObject` certainly suggests hidden complexity: forward declared, default value from code, read-write and read-only structs. All expressed to UHT as the same thing.

```
// ScriptStruct CoreUObject.GbxInlineStruct
// 0x0018 (0x0018 - 0x0000)
struct alignas(0x08) FGbxInlineStruct final
{
public:
        Pad_0[0x18];    // 0x0000(0x0018)(Fixing Struct Size After Last Property [ Dumper-7 ])
};

// Class GbxCore.TestInlineStructObject - I'm from GBX and I'm here to help
```

```
// /Script/OakGame.OakInteractiveObject
struct FGbxInlineStruct    DamageState;    // 0x1D68(0x0018)(Transient, NativeAccessSpecifierPrivate)
```

Also, this showed up with a confusing error message when used as a `TArray<GbxInlineStruct>`; the "`<unknown ArrayProperty>`" which is more correctly an ArrayProperty of an unsupported / unknown type.

```
In [89]: pc.RewardsManager.packages[0].RewardsDef._experimental_instance.reward
Out[89]: {rewarddata: <unknown ArrayProperty>, CustomData: <unknown GbxInlineStructProperty>, bIsUnic
In [93]: pc.RewardsManager.packages[0].RewardsDef._experimental_instance.reward.rewarddata._type
Out[93]: GbxInlineStructProperty'/Script/GbxGame.GbxRewardsSet:rewarddata.rewarddata'
```

`TMulticastInlineDelegate<>`, found in most (or all) of the hook-your-function-here parts of objects. Moderately useful, because they can point to functions that are triggered at an appropriate time to hook, but not directly useful unless we can add Python methods to them. eg:

```
// /Script/OakGame.OakCharacter
TMulticastInlineDelegate<void()>    OnStartedLadderSliding;    // 0x3DA0(0x0010)(ZeroConstructor, Ins
```

`FGameDataHandleProperty_`, as Dumper-7 identifies it, found all over the place. Points to other objects, often Actors or InteractableObjects. eg:

```
// /Script/OakGame.OakPlayerController
FGameDataHandleProperty_    ResurrectStation;    // 0x2388(0x0018)(Net, RepNotify, NoDestructor, hasG
```

Open **Unhandled types in BL4** #74

mattmab 4 days ago

Inspecting the Gilded Glory Rewards Package

In GbxRewardsManager

Under the Players Reward Manager (GbxRewardsManager_##########)

Packages.0.RewardsDef.rewarddata.0

FGbxDefPtr exists but is defined with FGbxDefPtr('None", None)

Here is an example of my console output cloning the reward package

```
glh_give_reward package 0 all <<<
GroundLootHelpers | GiveReward: using packages[0].RewardsDef =
FGbxDefPtr('RewardPackage_PreOrder', {...}) (all players)
GroundLootHelpers | GiveRewardPackageAllPlayers ptr TypeError:
GiveRewardPackageAllPlayers() missing 1 required positional argument: 'WorldContextObject'
GroundLootHelpers | GiveRewardPackageAllPlayers OK: ptr_pc
glh_give_reward package 0 <<<
GroundLootHelpers | GiveReward: using packages[0].RewardsDef =
FGbxDefPtr('RewardPackage_PreOrder', {...})
GroundLootHelpers | GiveReward OK: RewardDef_then_OwnerContext PC
```

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"In [89]: pc.RewardsManager.packages[0].RewardsDef._experimental_instance.reward
Out[89]: {rewarddata: <unknown ArrayProperty>, CustomData: <unknown GbxInlineStructProperty>, bIsUnique: False, UniqueName: 'Reward_ItemPool_LifesPurpose'}
In [93]: pc.RewardsManager.packages[0].RewardsDef._experimental_instance.reward.rewarddata._type
Out[93]: GbxInlineStructProperty'/Script/GbxGame.GbxRewardsSet:rewarddata.rewarddata'"

"TMulticastInlineDelegate<void()> OnStartedLadderSliding; // 0x3DA0(0x0010)(ZeroConstructor, InstancedReference, BlueprintAssignable, NativeAccessSpecifierPublic)"

"FGameDataHandleProperty_ ResurrectStation; // 0x2388(0x0018)(Net, RepNotify, NoDestructor, HasGetValueTypeHash, NativeAccessSpecifierPrivate)"

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